

Table 50. I/O current injection susceptibility

Symbol	Description		Functional susceptibility		Unit
			Negative injection	Positive injection	
I_{INJ}	Injected current on pin	Any IO	5 ⁽¹⁾	NA	mA

1. Evaluated by characterization. Not tested in production.

5.3.13 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in [Table 51](#) are derived from tests performed under the conditions summarized in [Table 24: General operating conditions](#). All I/Os are designed as CMOS- and TTL-compliant.

For information on GPIO configuration, refer to the application note AN4899 *STM32 GPIO configuration for hardware settings and low-power consumption*, available on the ST website www.st.com.

Table 51. I/O static characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$V_{IL}^{(1)}$	I/O input low level voltage	All	$1.65\text{ V} < V_{DDIOx} < 3.6\text{ V}$	-	-	$0.3 \times V_{DDIOx}$	V
$V_{IH}^{(1)}$	I/O input high level voltage	All	$1.65\text{ V} < V_{DDIOx} < 3.6\text{ V}$	$0.7 \times V_{DDIOx}$	-	-	V
$V_{hys}^{(2)}$	I/O input hysteresis	-		-	200	-	mV
$I_{Ikg}^{(3)}$	Input leakage current ⁽³⁾	FT_u	$0 < V_{IN} \leq V_{DDIOx}$	-	-	± 150	nA
			$V_{DDIOx} \leq V_{IN} \leq V_{DDIOx} + 1\text{ V}$	-	-	2500	
			$V_{DDIOx} + 1\text{ V} \leq V_{IN}$	-	-	250	
		Other FT_xx	$0 < V_{IN} \leq V_{DDIOx}$	-	-	± 70	
			$V_{DDIOx} \leq V_{IN} \leq V_{DDIOx} + 1\text{ V}$	-	-	600	
			$V_{DDIOx} + 1\text{ V} \leq V_{IN}$	-	-	150	
R_{PU}	Weak pull-up equivalent resistor ⁽⁴⁾	$V_{IN} = V_{SS}$		25	40	55	kΩ
R_{PD}	Weak pull-down equivalent resistor ⁽⁴⁾	$V_{IN} = V_{DDIOx}$		25	40	55	kΩ
C_{IO}	I/O pin capacitance	-		-	5	-	pF

1. Refer to [Figure 23: I/O input characteristics](#).

2. Specified by design. Not tested in production.

3. This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula: $I_{Total_leak_max} = 10\text{ }\mu\text{A} + [\text{number of I/Os where } V_{IN} \text{ is applied on the pad}] \times I_{Ikg}(\text{Max})$.